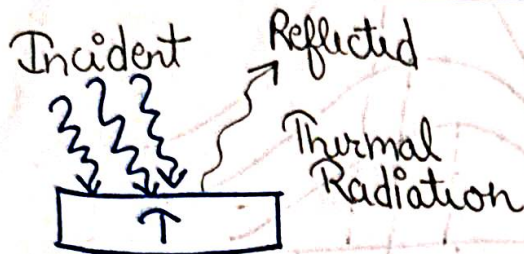


Quantum Physics

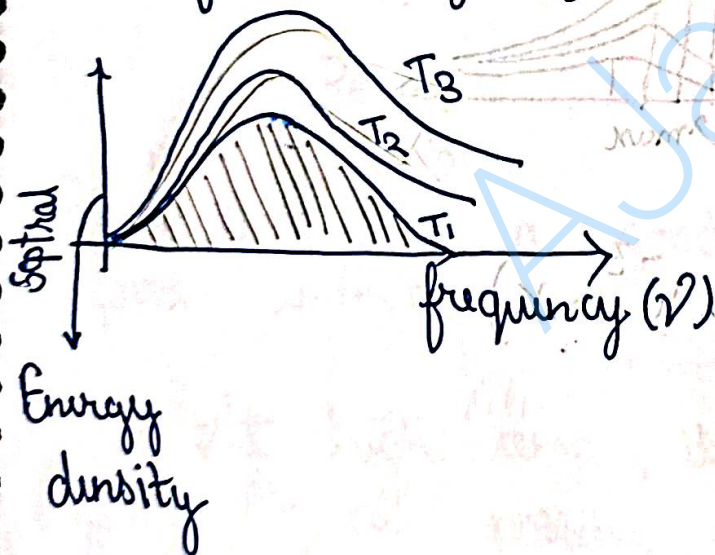
Black Body Radiation



Black body absorbs all types of radiations falling upon it.

Perfect absorber & a perfect emitter.

① Stefan-Boltzmann Law:



$$[T_1 < T_2 < T_3]$$

area under curve is Power

$$\Rightarrow P \propto T^4$$

$$P = \sigma T^4$$

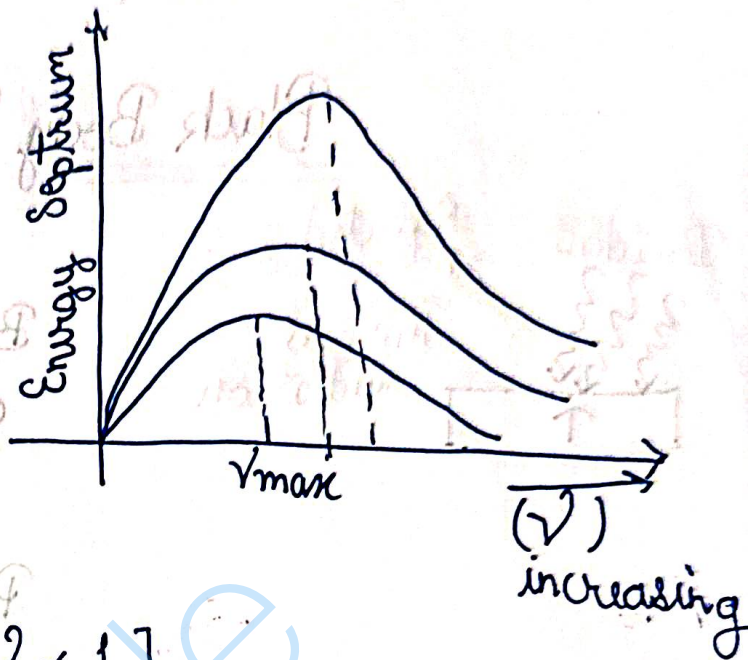
↓
Stefan-Boltzmann Constant

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

② Wien's displacement Law :

$$\nu_{\max} \propto T$$

Temp



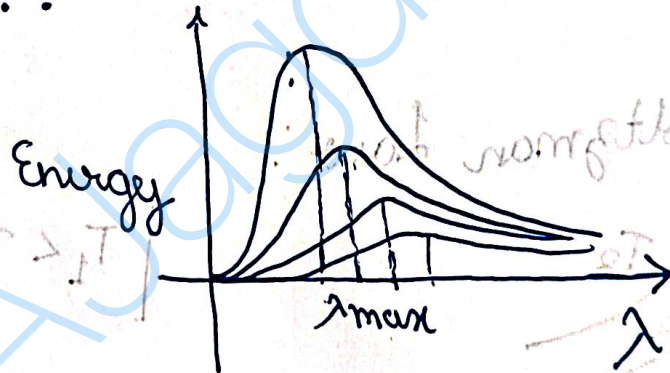
$$\nu_{\max} \propto T$$

$$\left[\nu \propto \frac{1}{\lambda} \right]$$

$$\frac{c}{\lambda_{\max}} \propto T$$

$\therefore$

$$c \propto \lambda_{\max} T$$



$$\therefore b = \lambda_{\max} T$$

where $(b = 2.89 \times 10^{-3} \text{ mK})$

$$T \propto \frac{1}{\lambda_{\max}}$$

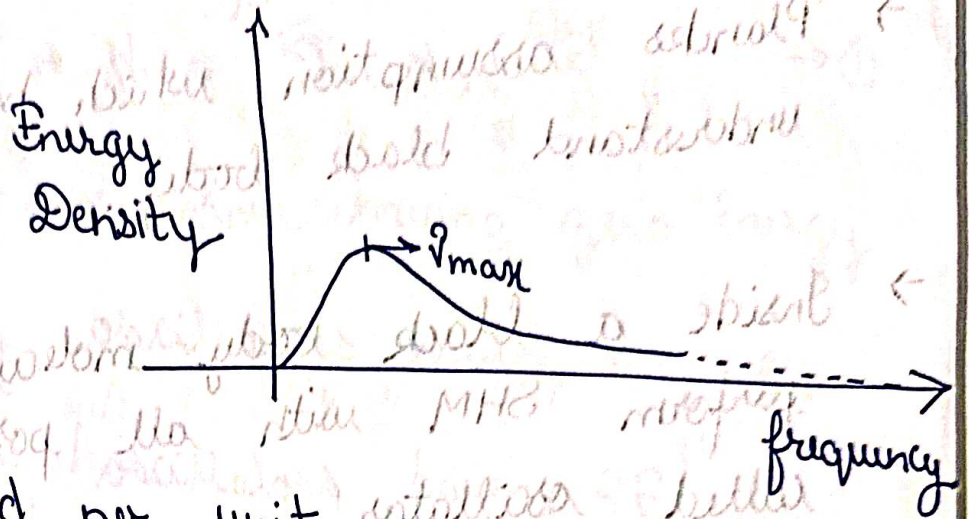
$$T \propto \frac{1}{\lambda_{\max}}$$

Step 1

Step 2

$$T \propto \frac{1}{\lambda_{\max}}$$

Black body



The energy emitted per unit time, per unit surface area for fixed interval of frequency like:

- is very small for low values of frequencies
- increases rapidly as frequency increases.

- Reflect
- Absorb
- emit.

Black body is a good absorber as well as a good emitter.

∴ At high temp, it will emit wavelengths of all possible.

and

At low temp, it will absorb radiations of all possible wavelengths.

Planck's Law

photon model

→ Planck's assumption which became law to understand black body

→ Inside a black body molecules vibrate and perform SHM with all possible frequencies called oscillator.

→ Oscillators cannot emit or absorb the energy continuously. But the energy of magnitude $h\nu$

$$E = h\nu$$

$$\nu = \frac{c}{\lambda}$$

and for n oscillators; $E = nh\nu$

Average Energy per oscillator.

$$N = N_0 + N_0 e^{-\frac{E}{Kt}} + N_0 e^{-\frac{2E}{Kt}} + N_0 e^{-\frac{3E}{Kt}} \rightarrow \textcircled{1}$$

$N_0 \rightarrow$ No. of oscillators having zero energy

$N_0 e^{-\frac{E}{Kt}} \rightarrow$ no. of oscillators having E energy.

$N_0 e^{-\frac{2E}{Kt}} \rightarrow$ no. of oscillators having $2E$ energy.

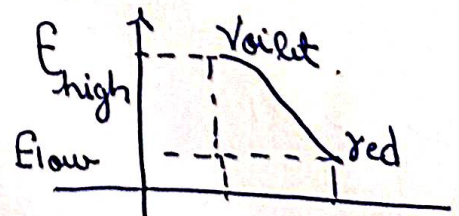
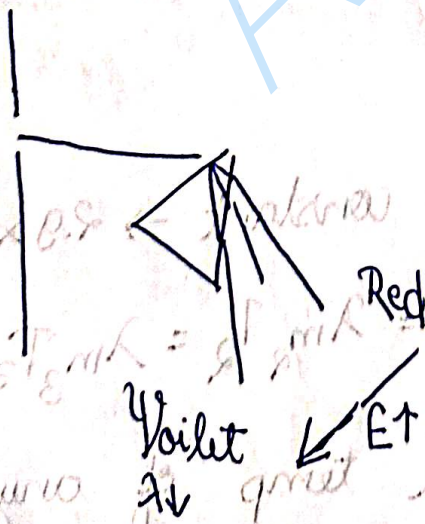
Put $x = \frac{E}{Kt}$ in eqⁿ $\textcircled{1}$

$$N = N_0 + N_0 e^{-x} + N_0 e^{-2x} + \dots$$

-ve represents that they are binded

$\rightarrow$ On heating iron rod, it changes its colour from red to white.

Black Body



Collimator

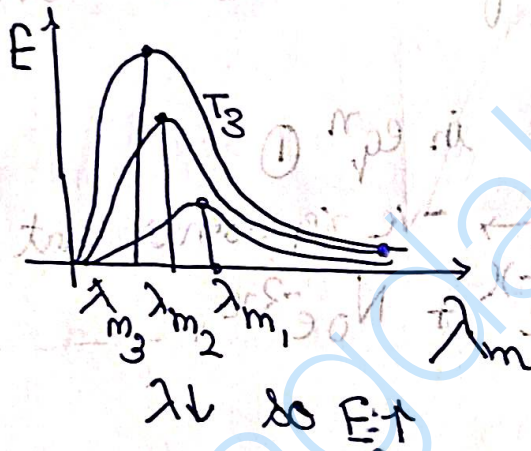
→ As we know on high frequencies; Energy will be more i.e.

$$E = \frac{hc}{\lambda}$$

as Temp ↑; Black Body is perfect emitter

∴ $E \uparrow$

so graphs on different temp.



$$T_3 > T_2 > T_1$$

We have different λ for different temp but
 $\lambda_m \times T \rightarrow \text{constant}$

$$\lambda_m T \rightarrow \text{constant} \rightarrow 2.9 \times 10^{-3} \frac{\text{m} \cdot \text{K}}{\text{meter}}$$

$$\lambda_{m1} \times T_1 = \lambda_{m2} \times T_2 = \lambda_{m3} \times T_3 = \text{const.}$$

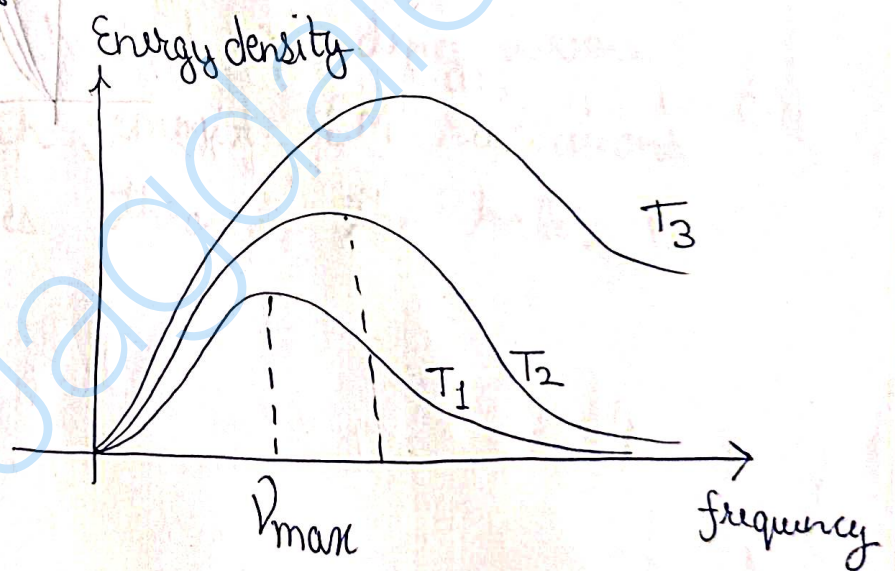
Used for calculating the temp of any body in space.

Quantum Mechanics:

- ① Rayleigh & Jean Law
- ② Weins displacement law
- ③ Stefan - Boltzman Law.

③ Stefan - Boltzman Law:

Power emitted by black body per unit time per unit surface area is proportional to the 4th power of temperature of the black body.



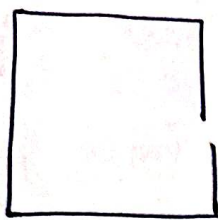
$$T_1 < T_2 < T_3$$

$$P \propto T^4$$

$$P = \sigma T^4$$

↓
Stefans - constant

Black Body :



Now when we pass light through the hole of cavity it reflects in the cavity and never comes back thru hole.

This means the "hole" in the cavity acts as a perfect black body.

When the cavity is heated standing waves are formed which emit EM radiations of all frequencies through the hole.

Black Body :

1) Rayleigh & Jeans

→ Number of modes per unit vol^m in frequency range ν to $(\nu + d\nu)$

$$\Rightarrow N = \frac{8\pi\nu^2}{c^3}$$

∴ Electromagnetic Energy density : $(u(\nu, t))$

$$\frac{8\pi\nu^2}{c^3} \langle E \rangle$$

$$k_B t$$

$$\therefore u(\nu, t) = \frac{8\pi\nu^2}{c^3} k_B t$$

But this eqn is only true for lower frequencies.

as

$$\text{for } \nu \rightarrow \infty; u(\nu, t) \rightarrow \infty$$

but in reality

$$\nu \rightarrow \infty, u(\nu, t) \rightarrow 0$$

This is ultraviolet catastrophe.

$$0 \leftarrow (h\nu) / k_B t$$

To resolve ultraviolet catastrophe

Max Planck assumed:

Energy is quantised
and electromagnetic
radiation will exchange
energy only in integral
multiples of $h\nu$

The avg energy can be
written as:

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} n h \nu e^{-\frac{n h \nu}{k_B T}}}{\sum_{n=0}^{\infty} e^{-\frac{n h \nu}{k_B T}}} = \left[\frac{h \nu}{e^{\frac{h \nu}{k_B T}} - 1} \right]$$

$$\therefore u(\nu, T) = \frac{8 \pi \nu^2 h \nu}{c^3 \left[e^{\frac{h \nu}{k_B T}} - 1 \right]}$$

$\therefore$ From above we can write

(a) For high frequencies:

$$\nu \rightarrow \infty$$
$$\therefore \frac{h \nu}{k_B T} \rightarrow \infty$$

$$\text{Resulting } u(\nu, T) \rightarrow 0$$

⑥ For lower frequencies

$\Rightarrow \frac{h\nu}{eK_Bt} - 1 =$ By Taylor series:

$$\left[1 + \frac{h\nu}{K_Bt} \right] - 1 = \left[\frac{h\nu}{K_Bt} \right]$$

so
$$u(\nu, t) = \frac{8\pi\nu^2}{c^3} K_Bt$$

Energy density
in different terms:

① In terms of wavelength (λ)

$$\boxed{\frac{c}{\lambda} = \nu}$$

$$\left| \frac{d\nu}{d\lambda} \right| = \frac{\frac{c}{\lambda}}{\lambda} \Rightarrow \left| \frac{c}{\lambda^2} \right|$$

so,
$$u(\nu, t) \cdot \left| \frac{d\nu}{d\lambda} \right| = u(\lambda, t)$$

$$\Rightarrow \frac{8\pi\nu^2}{c^3} \cdot \frac{h \left(\frac{c}{\lambda} \right)}{e^{\frac{hc}{\lambda K_Bt}} - 1} \cdot \frac{c}{\lambda^2}$$

$$u(\lambda, t) = \frac{8\pi}{c^3} \frac{c^2}{\lambda^2} \cdot \frac{1}{\lambda^3} \left[\frac{hc}{\left(e^{\frac{hc}{\lambda K_Bt}} - 1 \right)} \right]$$

$$\Rightarrow \frac{8\pi}{\lambda^5} \frac{hc}{\left[e^{\frac{hc}{\lambda K_Bt}} - 1 \right]}$$

⑥ In terms of angular frequency ω ⑦

$$\left| \frac{dv}{dw} \right| = \frac{v}{2\pi v} = \frac{1}{2\pi}$$

and

$$\Rightarrow \omega = 2\pi \nu$$

$$\Rightarrow \mu(\omega, t) = \frac{8\pi \left(\frac{\omega}{2\pi}\right)^2}{c^3} \cdot \left[\frac{h \left(\frac{\omega}{2\pi}\right)}{e^{\frac{h(\omega)}{2\pi k_B t}} - 1} \right] \cdot \frac{1}{2\pi}$$

$$u(\omega, t) \Rightarrow \frac{h\omega^3}{\pi^2 c^3} \frac{1}{\left[e^{\frac{h\omega}{k_B t}} - 1 \right]}$$

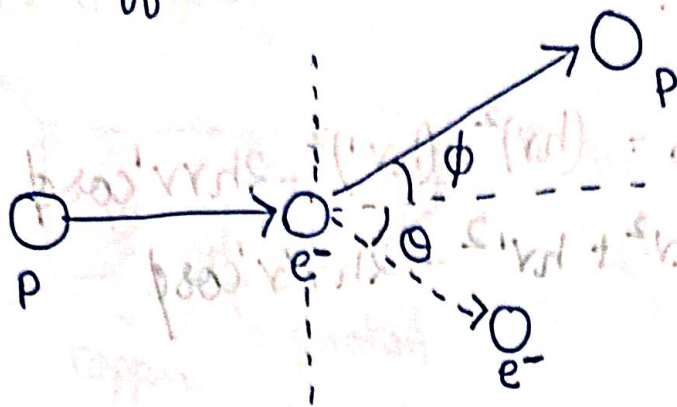
Wien's displacement law

$\Rightarrow \lambda_{\text{max}} T = b$
 $\downarrow$
 Wien's
 Constant
 $2.8 \times 10^{-3} \text{ m} \cdot \text{K}$

$$\left[\frac{1}{1 - \frac{\frac{2N}{\pi}}{2\pi N}} \right] \frac{1}{\pi} \cdot \frac{2\pi}{\pi} = (f_{\text{eff}})_{\text{eff}}$$

$$\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Compton Effect



Let us assume energy of photon having energy $= h\nu$ and energy $(s) = h\nu'$

$\therefore$ momentum can be written $= \frac{h\nu}{c}$

applying momentum conservation:

$$\text{In } x : \frac{h\nu}{c} = \frac{h\nu' \cos \phi}{c} + p \cos \theta$$

$$\therefore h\nu - h\nu' \cos \phi = pc \cos \theta \Rightarrow (1)$$

$$\text{In } y : 0 = \left[-\frac{h\nu' \sin \phi}{c} \right] + p \sin \theta$$

$$\Rightarrow \frac{h\nu' \sin \phi}{c} = p \sin \theta \Rightarrow (2)$$

$$h\nu \sin \theta = pc \sin \theta$$

On squaring & adding:

$$\Rightarrow \underbrace{p^2 c^2 \cos^2 \theta + p^2 c^2 \sin^2 \theta}_{p^2 c^2 = E^2} = (h\nu)^2 + (h\nu')^2 - 2h\nu\nu' \cos \phi$$

Total Energy of Particle

$$\Rightarrow E_{\text{Total}} = KE + m_0 c^2$$

$\downarrow$
 Rest Energy

we get

$$E_{\text{Total}}^2 = p^2 c^2 + m_0^2 c^4$$

$$\therefore E = KE + mc^2$$

&

$$E = p^2 c^2 + m_0^2 c^4$$

$$\& KE = h\nu - h\nu'$$

from all these ;

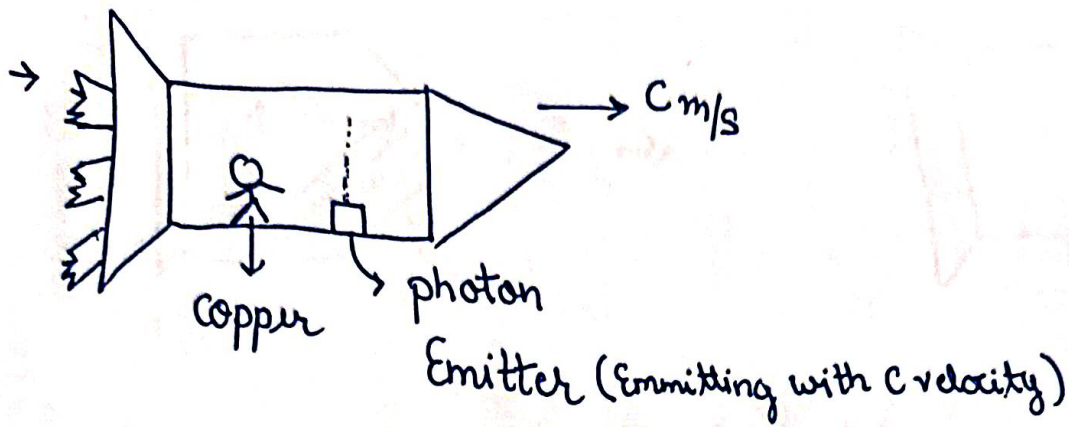
$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos \phi)$$

$\downarrow$
 λ_c

$\rightarrow$ Compton wavelength

$$\lambda_c = 2.426 \times 10^{-12} \text{ meter}$$

Time dilation:



line

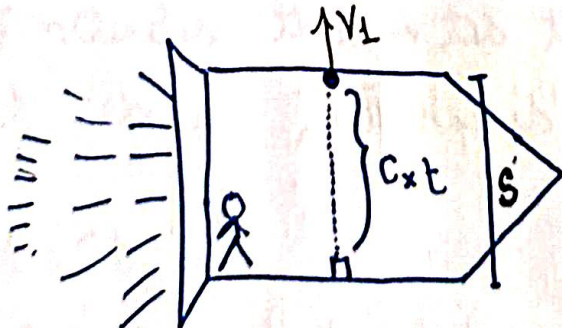
Means;

Let us say the rocket moves with velocity $v = c \text{ m/s}$ i.e. speed of light and anything cannot go faster than "c".

Now The observer Copper is in the rocket & line outside.

For an interval, both see or observe the photon emitted

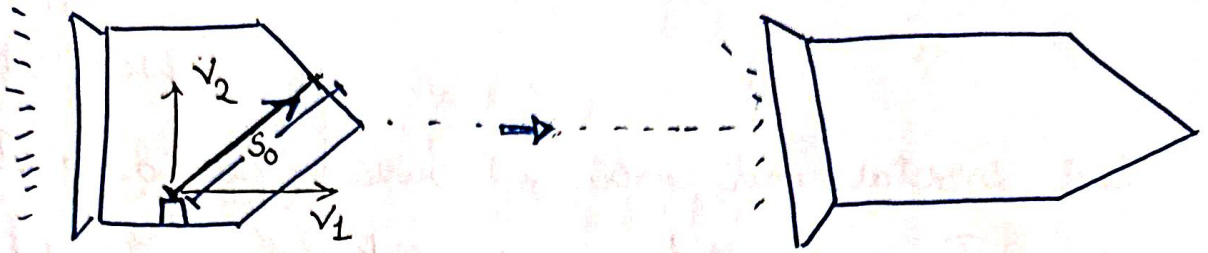
POV Copper →



$$S = c \times t$$

↓
distance

Line POV



Line

here we can clearly see the $S_0 > S$ (distance)
Travelled by Photon.

But; both observe for the same interval. of ~~time~~

$$S = ct$$

$$S_0 = ct$$

} → This cannot be true

∴ This concludes that → the speed of photon
cannot change; There is a "TIME DILUTION"

⇒

$$S = ct$$

and

$$S_0 = ct_0$$

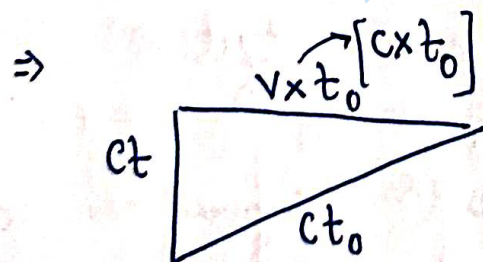
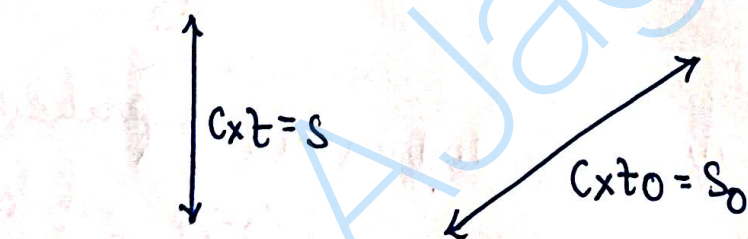
where $[t_0 > t]$

TIME DILUTION

→ Both observers observe for same time interval but the time is different in both case. This is "time dilation" discovered by Albert Einstein

Explaining how time can pass at different rates for observer depending on their relative speed or presence in gravitation field.

Further we can find the values as



$v \rightarrow$ velocity of rocket

$\Rightarrow$

$$(ct_0)^2 = (ct)^2 + (vt_0)^2$$

$$\Rightarrow c^2 t_0^2 = c^2 t^2 + v^2 t_0^2$$

$$\Rightarrow c^2 t_0^2 - v^2 t_0^2 = c^2 t^2$$

$$\Rightarrow \frac{c^2 t_0^2 - v^2 t_0^2}{c^2} = t^2$$

SMIT
NOTION

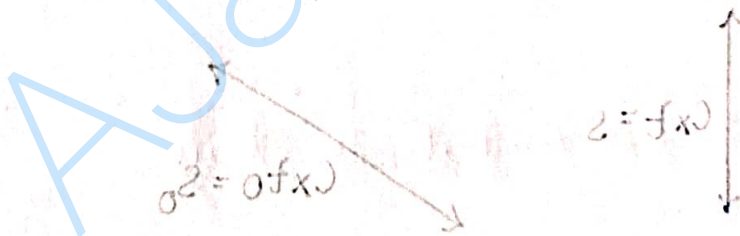
but $t_0^2 - \frac{v^2}{c^2} t_0^2 = t^2$ of moving observer at
at rest, so c^2 is sufficient at rest at

$$\Rightarrow t_0^2 \left(1 - \frac{v^2}{c^2}\right) = t^2$$

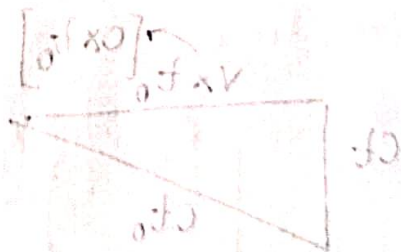
travelling to stop and with each principle
observer will $t_0 = \frac{t}{\sqrt{1 - \frac{v^2}{c^2}}}$ [Internal
observer time observed] both

[External
observer
time
observed]

author will bring out the result



author's perspective



$$f_0(V) = f(t) = f_0(t)$$

$$f_0(t) + f_0(t) = f_0(t)$$

$$f_0(t) = f_0(t) - f_0(t)$$

At higher velocities: $v \rightarrow c$

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{ie } E = mc^2 \cdot \gamma$$

and

$$p = \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}} \rightarrow p = (mv \cdot \gamma)$$

For to Prove $E^2 = p^2 c^2 + m_0^2 c^4$

we have to

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Squaring

$$\Rightarrow E^2 = \frac{m^2 c^4}{1 - \frac{v^2}{c^2}}$$

$\Rightarrow$ Inserting $(v^2 - v^2)$ gives

$$\Rightarrow E^2 = \frac{m^2 (v^2 - v^2 + c^4)}{1 - \frac{v^2}{c^2}}$$

$$\Rightarrow E^2 = \frac{m^2 v^2}{1 - \frac{v^2}{c^2}} - \frac{m^2 v^2}{1 - \frac{v^2}{c^2}} + \frac{m^2 c^4}{1 - \frac{v^2}{c^2}}$$

$$\Rightarrow E^2 = \left(\frac{mv}{\sqrt{1-\frac{v^2}{c^2}}} \right)^2 c^2 + \frac{m^2 c^4 - m^2 c^2 v^2}{1-\frac{v^2}{c^2}}$$

$$\Rightarrow E^2 = p^2 c^2 + m^2 c^2 \left(\frac{c^2 - v^2}{1-\frac{v^2}{c^2}} \right)$$

$$\Rightarrow E^2 = p^2 c^2 + m^2 c^2 \left(\frac{c^2 - v^2}{\frac{c^2 - v^2}{c^2}} \right)$$

$$\Rightarrow E^2 = p^2 c^2 + m^2 c^2 \cdot c^2$$

Hence proved

De Broglie Wave phase and group velocities

as $\frac{h}{\lambda} = p$

$\therefore \frac{h}{p} = \lambda$

$\therefore$ DeBroglie wavelength λ for a particle having rest mass m_0

$\Rightarrow \frac{h}{\gamma m_0 v}$

$\gamma = \frac{1}{\sqrt{1 + \frac{v^2}{c^2}}}$

Phase Velocity : The velocity with which the deBroglie wave travels is called phase velocity.

is defined by

$v_p = \lambda \nu$

Identity

$\nu \rightarrow$ frequency

$\lambda \rightarrow$ wavelength

We know;

$$\lambda = \frac{h}{p} = \frac{h}{(\gamma m_0 v)}$$

By mass energy Relation

$$E = h\nu = \gamma m_0 c^2$$

$$\Rightarrow \nu = \frac{\gamma m_0 c^2}{h}$$

$\therefore$

$$v_p = \nu \times \lambda$$

$\Rightarrow$

$$\frac{\gamma m_0 c^2}{h} \times \frac{h}{(\gamma m_0 v)}$$

$$\left[\frac{c^2}{v} \right]$$

Here we get;

Velocity of de-Broglie wave is faster than the velocity of particle as $v \ll c$.

But;

Since the debroglie wave is associated with a moving body; one can expect that the wave should

Velocity as the particle.

further since $v \ll c$ so $v_p \gg c$, which means de-Broglie wave travels much faster than the velocity of light. which is wrong.

Group Velocity (v_g)

Now due to failure of phase velocity, we introduce group velocity.

$$v_g = \frac{d\omega}{dk}$$

$$\text{Since } \omega = 2\pi\nu$$

$$\omega = 2\pi \gamma \frac{m_0 c^2}{h}$$

$$\text{and } k = \frac{2\pi f}{\lambda} \text{ or } \frac{2\pi}{\lambda} \rightarrow \frac{2\pi m_0 v}{h(1 - \frac{v^2}{c^2})^2}$$

$$\therefore \underline{v_g = v}$$

De Broglie wave group associated with the moving particle travel with the same velocity as the particle.

Heisenberg Uncertainty Principle

$$\Delta x \cdot \Delta p \geq \frac{h}{2}$$

Derivation (1)

$$\frac{dx}{dt} = v$$

$$v = \frac{h}{m\lambda}$$

$$m = \frac{h}{\lambda v}$$

$$\lambda = \frac{h}{m v} = \frac{h}{\frac{h}{\lambda v} v} = \lambda$$

$$v = \frac{h}{m\lambda}$$

So we have now shown that the product of the uncertainties in position and momentum is greater than or equal to h/2.

Operators in Quantum Mechanics:

Operators act on a function and produce another function.

$$f(x) \longrightarrow \boxed{\hat{A}} \longrightarrow g(x)$$

Operators must satisfy two conditions:

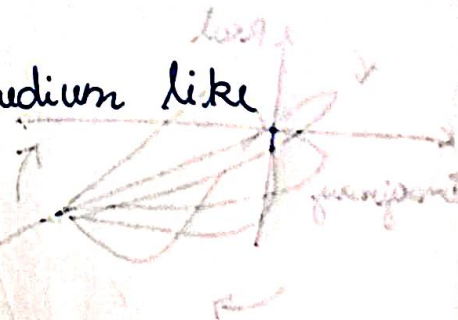
$$(i) \hat{A}(af(x)) = a \hat{A}f(x)$$

$$(ii) \hat{A}(f(x) + g(x)) = \hat{A}f(x) + \hat{A}g(x)$$

Quantum Waves

→ are not real waves; There is no medium like water or air.

we can't measure them directly.



Wave function

$$\Psi \rightarrow (\psi, i)$$

operations are done on wave function to find
probability in position
probability in momentum
probability in energy
etc.

Just probability amplitude
Keeps track of properties of quantum particle.

Wavefunction Constraints:

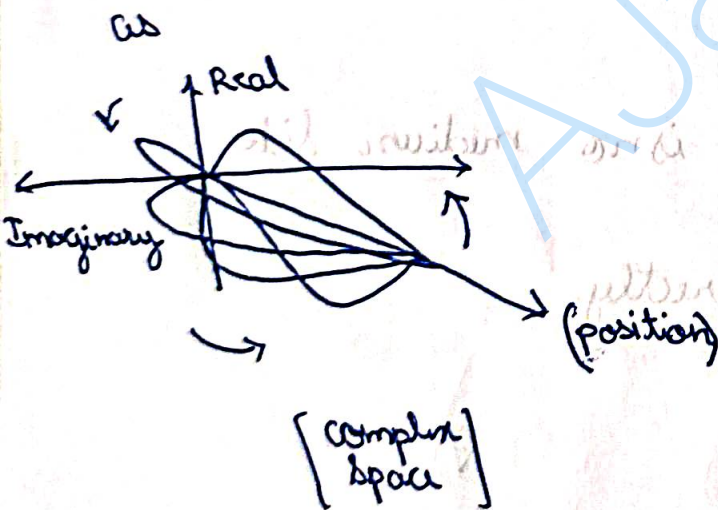
⇒ ① Satisfy the Schrodinger Eqⁿ:

$$i\hbar \frac{d\psi}{dt} = \frac{-\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V \cdot \psi$$

⇒ ② Normalisable

means the wavefunction when multiplied by its conjugate give [Probability distribution].

The area under the curve should be finite
 bcz there has to be a definite probability
 you will find the particle somewhere.



ψ is the wave function

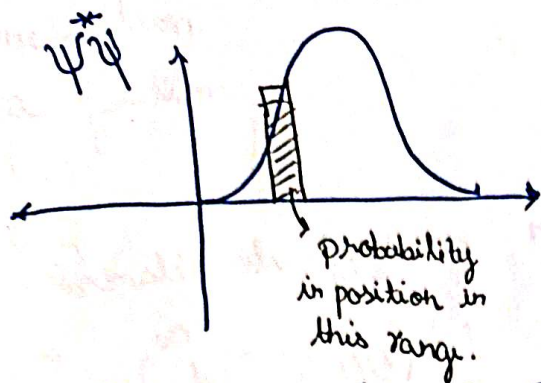
ie
 $\psi = a + ib$

Now

$\psi^* \rightarrow$ conjugate
 ie $\rightarrow a - ib$

So $\psi^* \psi \rightarrow$ gives real part.

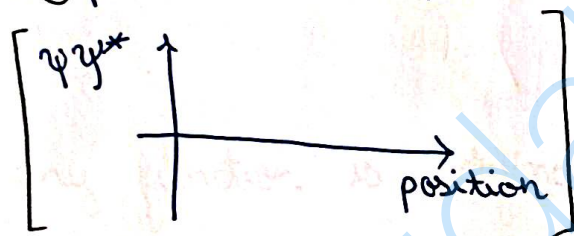
and this is the "Probability Density of a Probability Distribution."



(position, momentum, energy) ^{quantities} (1)
 (trajectories) (2)

as Probability ≤ 1 $\therefore$ It means The area under the curve should be finite.

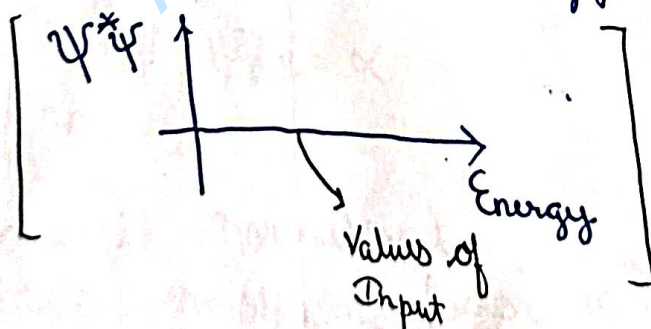
$\therefore$ In quantum physics we always get probabilities
 i.e. ① probabilities in position



② probabilities in momentum



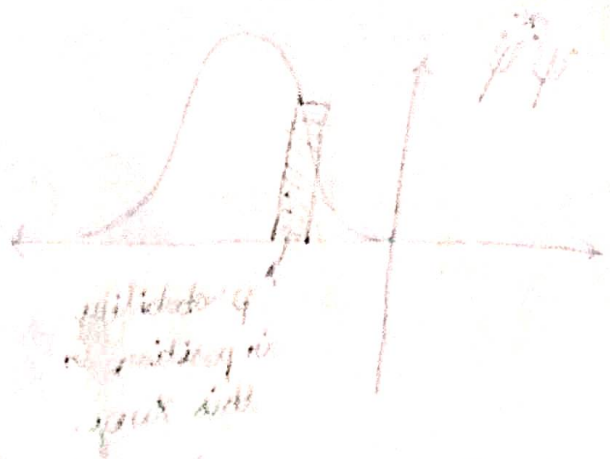
③ probabilities in Energy



③ Single valued

④ Continuous

⑤ Continuous gradient.



with continuous wave function, wave function is continuous and its gradient is also continuous.

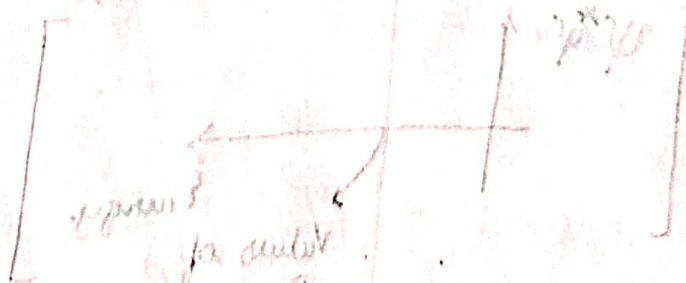
Probability density function is continuous and its gradient is also continuous.



Probability density function is continuous



Probability density function is continuous



Normalisation of a function:

$\therefore$ Normalisable function means

$$\Rightarrow \int_{-\infty}^{\infty} \psi^* \psi d(x) = 1$$

↓
Nothing complex only says the probability density or area under curve = 1

Now

If any function is not normalisable
ie

$$\int_{-\infty}^{\infty} \psi^* \psi dx \neq 1$$

ie ψ is not normalisable $\Rightarrow$ ψ is not a wave function

$$\int_{-\infty}^{\infty} \psi^* \psi dx = \alpha$$

So to make it normalised
we add a normalizing factor ie "N"

$N\psi$ = New wave function ie ψ_1

For value of "N"

$$\int_{-\infty}^{\infty} N^2 \psi^* \psi dx = 1$$

$$N^2 \int_{-\infty}^{\infty} \psi^* \psi dx = 1$$

$$N^2 = 1$$

$$N = \frac{1}{\sqrt{\alpha}}$$

$$\therefore \psi_1 = \frac{1}{\sqrt{\alpha}} \psi$$

$\Rightarrow$ where ψ_1 is normalised form of ψ .

Defining Eigenvalues
and Eigenvectors:

$$A = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A\vec{x} = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

$$\Rightarrow A\vec{x} = -2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

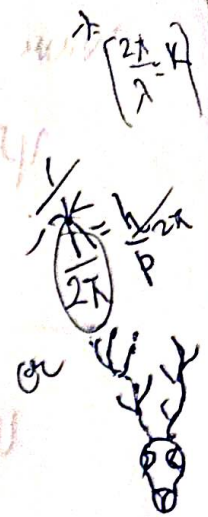
where $\vec{x}$ is an eigenvector
and -2 is an eigenvalue.

$$\therefore A\vec{x} = \lambda\vec{x}$$

$$\therefore A\vec{x} = \lambda I\vec{x}$$

$$\Rightarrow A\vec{x} - \lambda I\vec{x} = 0$$

$$(A - \lambda I)\vec{x} = 0$$



$$\left(\frac{1}{\omega} - \frac{1}{\omega} \right) = 0$$

$$\left(\frac{1}{\omega} - \frac{1}{\omega} \right) = 0$$

$$\begin{bmatrix} \omega & \omega \\ \omega & \omega \end{bmatrix}$$

Wave function

ψ & $\psi \psi^* \rightarrow$ gives probability distribution

$$\psi = A e^{-i(\omega t - kx)} \rightarrow \textcircled{1}$$

where $\left(\omega = \frac{E}{\hbar} \text{ sec}^{-1} \right)$

$$\left(k = \frac{p}{\hbar} \text{ m}^{-1} \right)$$

Schrodinger Eqⁿ:

$$\psi = A e^{-i\left(\frac{E}{\hbar} t - kx\right)}$$

$$\omega = 2\pi \nu$$

$$h\nu = E$$

$$\nu = \frac{E}{h}$$

$$\omega = 2\pi \frac{E}{h}$$

$$\boxed{\omega = \frac{E}{\hbar}}$$

$$k = \frac{2\pi}{\lambda}$$

$$\Rightarrow \frac{h\nu}{\lambda} = E$$

$$\lambda = \frac{hc}{E}$$

$$k = \frac{2\pi E}{hc}$$

$$\frac{h}{\lambda} = p$$

$$\therefore k = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{h}{p}$$

$$\therefore k = \frac{2\pi \cdot p}{h}$$

$$\text{or } \frac{p}{\hbar}$$

$$\psi = A e^{-i \left[\frac{Et}{\hbar} - \frac{Px}{\hbar} \right]}$$

$$\psi = A e^{-\frac{i}{\hbar} [Et - Px]}$$

$$\frac{d\psi}{dx} = A \frac{iP}{\hbar} e^{-\frac{i}{\hbar} (Et - Px)}$$

$$\frac{d^2\psi}{dx^2} = A \left(\frac{iP}{\hbar} \right)^2 e^{-\frac{i}{\hbar} (Et - Px)}$$

$$\frac{d^2\psi}{dx^2} = \left(\frac{iP}{\hbar} \right)^2 A e^{-\frac{i}{\hbar} (Et - Px)}$$

$$\Rightarrow \frac{d^2\psi}{dx^2} = P^2 \left(\frac{1}{\hbar} \right)^2 \psi$$

$$\Rightarrow - \left[\frac{P}{\hbar} \right]^2 \psi = \frac{d^2\psi}{dx^2}$$

$$\boxed{P^2 \psi = -\hbar^2 \frac{d^2\psi}{dx^2}} \rightarrow (4)$$

This is when
we double diff
wrt dx

Then

$$\psi = A e^{-i(\frac{E}{\hbar}t - \frac{p}{\hbar}x)}$$

$$\Rightarrow \frac{d\psi}{dt} = -A \left[i \frac{E}{\hbar} \right] e^{-i(\frac{E}{\hbar}t - \frac{p}{\hbar}x)}$$

$$\Rightarrow \frac{d\psi}{dt} = -\frac{iE\psi}{\hbar}$$

$$\boxed{E\psi = \frac{\hbar}{i} \frac{d\psi}{dt}} \rightarrow \text{diff single time wrt } t$$

Now Total Energy of a Particle

$$\Rightarrow KE + P.E$$

$$E = \frac{1}{2}mv^2 + V(x,t)$$

$$\textcircled{11} \quad E = \frac{p^2}{2m} + V(x,t)$$

multiplying by ψ

$$\Rightarrow E\psi = \frac{p^2}{2m}\psi + V\psi(x,t)$$

$$i\hbar \frac{d\psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi \rightarrow (9)$$

as $E\psi$ & $\frac{p^2}{2m}\psi$ are derived previously.

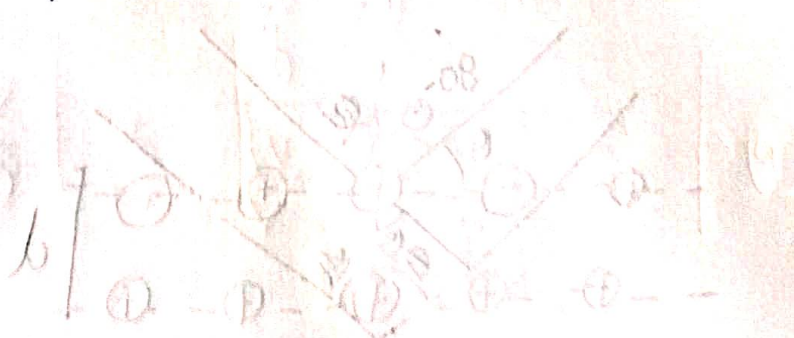
$$\Rightarrow i\hbar \frac{d\psi}{dt} = -\frac{\hbar^2}{2m} \left[\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} \right] + V\psi$$

&

time independent Schrodinger Eq $\Rightarrow$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

here, $\psi = A e^{i p x / \hbar}$



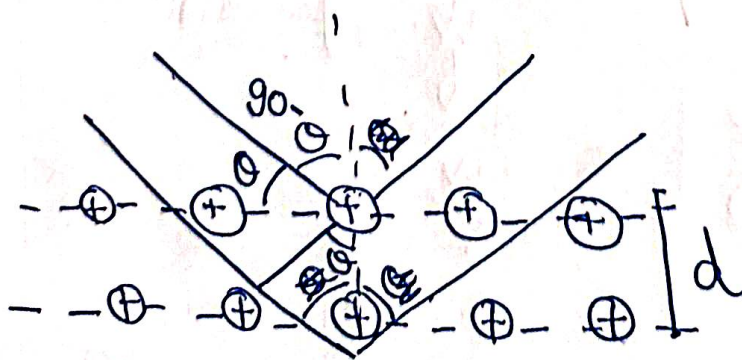
Wave nature:

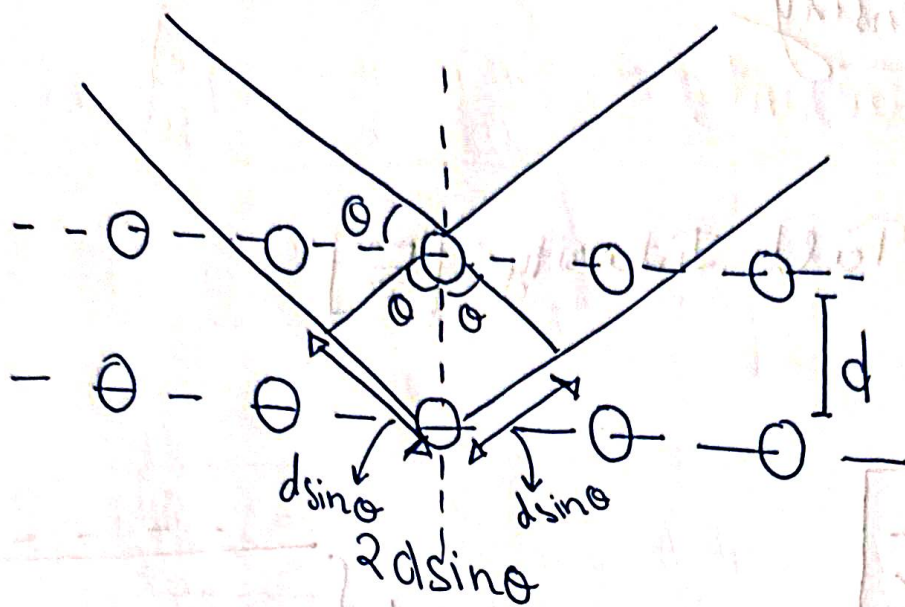
When Ni-foil was heated on bombarding electron on it scattered intensely at a specific angle ϕ , which dependent on the initial KE of bombard electrons.

→ This conculed that Germer was observing wave character of electron proposed by de-Broglie.

→ This was bec Ni got crystallized; & got arranged in regular fashion.

This arranged the atoms; such that it reflects X-rays in a very specific direction.

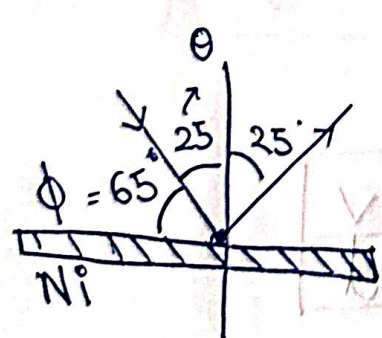




the difference between path lengths
travelled by the waves

$$\frac{W}{V} = \frac{h}{p} \Rightarrow \left[2d \sin \theta = n \lambda \right]$$

Incident photon wavelength



$$\lambda = \frac{h}{p}$$

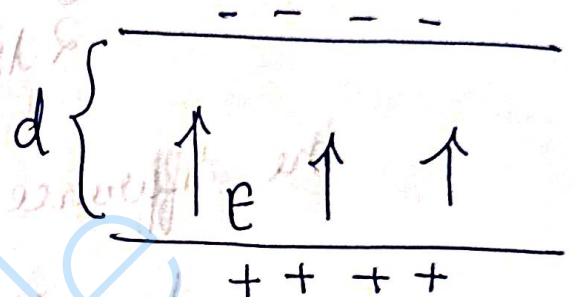
$$p = \sqrt{2mK.E}$$

Current Density

⇒ Electric Field Intensity: $[E]$

$$qE = F$$

$$E = \frac{F}{q}$$



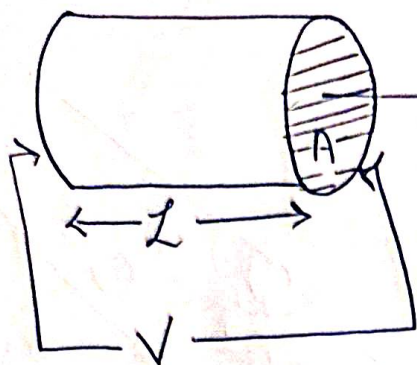
Now we have

$$\textcircled{1} \text{ Work Done} = V \cdot q \Rightarrow W = V \cdot q ; q = \frac{W}{V}$$

and

$$\text{Work done} = F \times d \Rightarrow W = F \times d ; F = \frac{W}{d}$$

$$E = \frac{\frac{W}{d}}{\frac{W}{V}} = \boxed{\frac{V}{d}}$$



known ρ & L
 find R

$$R = \rho \frac{L}{A}$$

Q

$$\sigma = \frac{1}{\rho}$$

conductivity

Microscopic or point form

$$E \cdot \sigma = J$$